

Feedback

Workshop 3

Workshop 3

- The third workshop looked at calculating numerical solutions to the 1D advection equation
- The specific areas we looked at were:
 - Changing the number of grid points
 - Changing the time step
 - Choosing a finite difference expression

Question 1:

Question 1: What is the end time of the calculation and how far has the Gaussian advected?

```
> ./advection1D
Grid spacing      = 0.0001
CFL number        = 0.9
Time step         = 0.009
No. of time steps = 5000
End time          = 45
Distance advected = 0.45
> █
```

End time of the calculation: $t=45$

Distance advected: $d=0.45$

Velocity: $v=0.01$

Which is what we expect from $d = vt$

Question 2:

- Reduce the number of grid points by a factor of 100 from 10 000 to 100. Then recompile and re-run the program.

Question 2: How does this affect the end time of the calculation and why?

NX=10000

```
> ./advection1D
Grid spacing      = 0.0001
CFL number        = 0.9
Time step         = 0.009
No. of time steps = 5000
End time          = 45
Distance advected = 0.45
> █
```

NX=100

```
> ./advection1D
Grid spacing      = 0.01
CFL number        = 0.9
Time step         = 0.9
No. of time steps = 5000
End time          = 4500
Distance advected = 45
> █
```

The end time is now 100 times larger.

$$\Delta t = C \frac{\Delta x}{v} \Rightarrow \Delta t \propto \Delta x$$

So when we increase the spacing of the grid points the time step increases in proportion.

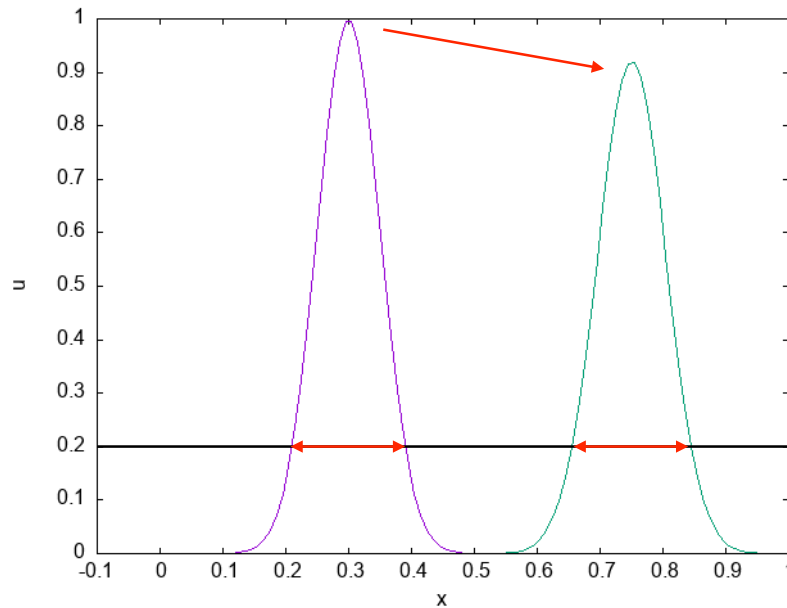
We only need 50 time steps to reach the same end time

Question 3:

- Change the number of time steps so that the end time is the same as your answer to question 1. Then recompile and re-run the program, and plot the new results.

Question 3: How has the reducing the resolution affected the accuracy of the solution?

$NX=100$, $nsteps=50$

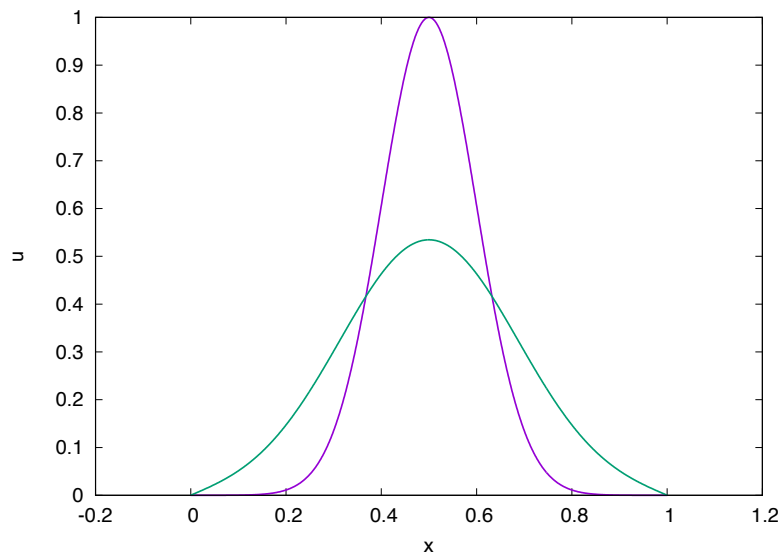


Smaller amplitude
Larger width

Hmm, haven't we
seen this before?

Question 3:

From Unit 3.4: The diffusion equation



The solution is behaving as if there is diffusion! This is a numerical effect (numerical diffusion) and it comes from the truncation error.

Increasing resolution is not just about having more detail. The resolution affects the quality of the solution more fundamentally.

Question 4:

Question 4: How has reducing the resolution changed the computational cost of calculating the solution? To answer this question think about how many calculations need to be performed.

NX: 10000 → 100 (100 times smaller)

nsteps: 5000 → 50 (100 times smaller)

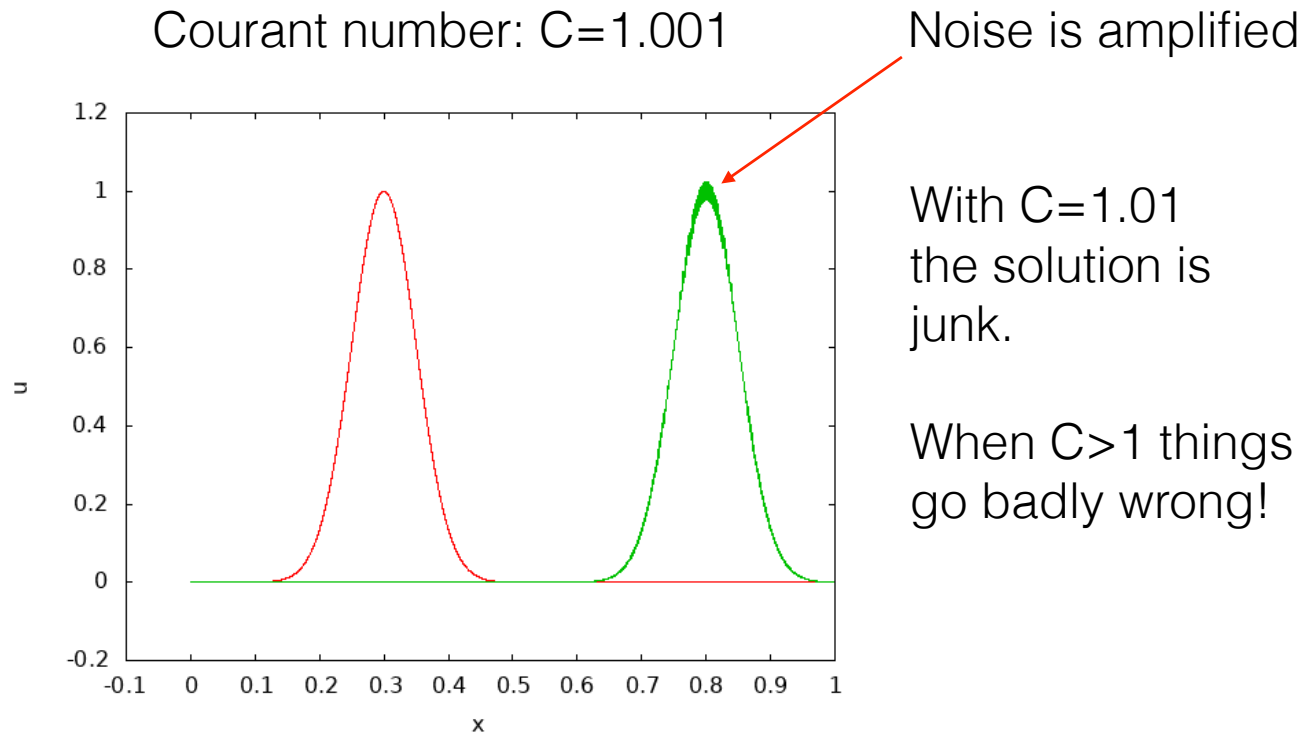
Overall the computational cost is 10 000 times less

Improving the spatial resolution is more computationally expensive than it appears

Worse in two or three dimensions!

Question 5:

Question 5: What happens if the CFL condition is not respected?



Question 6:

- Change the velocity (`velx`) to `-0.01`
- Now compile and run the program. This will not work correctly!

Question 6: What are the values of `u` in the file `final.dat`?

```
0.017450 inf
0.017550 -inf
0.017650 inf
0.017750 -inf
0.017850 inf
0.017950 -inf
0.018050 inf
0.018150 -inf
0.018250 inf
0.018350 -inf
0.018450 inf
0.018550 -inf
0.018650 inf
0.018750 -inf
0.018850 inf
0.018950 -nan
0.019050 -nan
0.019150 -nan
0.019250 -nan
0.019350 -nan
0.019450 -nan
0.019550 -nan
0.019650 -nan
0.019750 -nan
0.019850 -nan
0.019950 -nan
0.020050 -nan
0.020150 -nan
```

Infinites and NaNs (not a number)
The calculation has gone badly wrong!

Question 7:

Question 7: How would you modify the program to allow for advection in either direction (positive or negative velocities)?

- If `velx` is positive use the left-wards difference
- If `velx` is negative use the right-wards difference
- The finite difference must be taken on the upstream side
- This is known as an “up wind” scheme

Workshop 3 key points

- Increasing the spatial resolution reduces numerical errors (not just about having more “detail”)
- Increasing spatial resolution is computationally expensive as we have to decrease the time step
- Numerical schemes can go wrong for reasons other than bugs in the program:
 - Stability constraint not respected
 - Numerical scheme is unstable e.g. inappropriate choice of finite difference